Guillem Ramírez

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EDUCATION

University of Edinburgh (UoE) Sep. 2022 – Aug 2027
PhD in Natural Language Processing. Supervised by Prof. Ivan Titov Edinburgh, UK
Expected graduation: August 2027 (flexible). Available for internships March-August 2027
Researching efficient LLM systems through collaboration between models of different capabilities. Developed methods for uncertainty-based routing, selective knowledge distillation, and privacy-preserving interaction with external LLMs. Currently working on latent communication between models.

University of Edinburgh (UoE) Sep. 2020 – Sep. 2021
Masters of Science in Artificial Intelligence (with distinction) Edinburgh, UK

Massachusetts Institute of Technology (MIT) Sep. 2019 – July 2020
Final year of BSc at Prof. Marin Soljačić's group Cambridge, USA
Defended my Bachelor's thesis (with distinction).

Technical University of Catalonia (UPC) Sep. 2015 – May 2020
Dual Bachelor of Science in Mathematics and Engineering Physics Barcelona, Spain
Double degree coordinated by the Center of Superior Interdisciplinary Formation (CFIS) at UPC, a five-year excellence program offering two independent degrees.

SELECTED PUBLICATIONS

Improving Cross-Lingual Token Representations by Adding a Pinch of SALT (2026)
Guillem Ramírez. Paper
Developed a lightweight post-training method using span-level supervision that improves token-level cross-lingual representations in 4 out of 5 tasks while also enhancing sentence-level performance.

Controlling What You Share: Assessing Language Model Adherence to Privacy Preferences (2026)
Guillem Ramírez, Alexandra Birch, Ivan Titov. Accepted in **ACL Findings 2026**. Paper
Trained a lightweight model (8B) that rewrites queries according to user privacy preferences before external LLM inference.

Omnilingual SONAR: Cross-Lingual and Cross-Modal Sentence Embeddings (2026)
JM Janeiro, PL Huguet, I Tsiamas, Y Meng, V Iyer, **G Ramírez**, L Barrault et al. Paper

Optimising Calls to Large Language Models with Uncertainty-Based Two-Tier Selection (2024)
Guillem Ramírez, Alexandra Birch, Ivan Titov. **COLM 2024**. Paper. 40+ citations
Our uncertainty-based routing method outperforms existing methods in 25 of 27 experimental setups across 9 tasks.

Cache & Distil: Optimising API Calls to Large Language Models (2024)
Guillem Ramírez, Matthias Lindemann, Alexandra Birch, Ivan Titov. **ACL Findings 2024**. Paper. 30+ citations
An adaptive system where a small LLM handles easy queries while the difficult ones are delegated to a larger LLM; knowledge distillation progressively improves the small model, allowing it to replace more expensive LLM calls over time.

WORK EXPERIENCE

Research Scientist intern at FAIR (Meta) Aug. 2025 – Feb. 2026
Developed OmniSONAR-Token, a token-level extension of OmniSONAR covering 57 languages. Paris, France
Improved cross-lingual token alignment and achieved state-of-the-art performance on word alignment and sequence tagging while preserving sentence-level representation quality.

Applied Scientist intern at Amazon Aug. 2024 – Jan. 2025
Developed uncertainty-based cascading for LLM-as-a-judge evaluation, reducing evaluation costs by 20–70% while maintaining or improving accuracy across multiple benchmarks. Developed lightweight surrogate models using internal representations of smaller LLM judges to predict when to defer to larger models. Berlin, Germany

ML intern at Algomo Jun. 2022 – Sep. 2022
Building a Dutch summarisation model in collaboration with VRT. London, UK

Research Assistant at the University of Edinburgh Sep. 2021 – June 2022
Researching controlled text generation for data augmentation and machine translation. Edinburgh, UK
Adapting Machine Translation pipelines to a terminology list from the health domain. This involved fine-tuning and dockerising models for the BBC. Project lied within the GoURMET project of the European Union.

Data scientist intern at Base Technology & Information Services Jun. 2018 – Sep. 2018
Automatization of the extraction of information from CVs. Migrating the company's database and creating different indicators (SQL queries) of the company that are automatically updated on Data Studio. Barcelona, Spain

TECHNICAL SKILLS

Programming Python, C++, Bash, SQL

ML Pytorch, Tensorflow, Fairseq, FlashAttention, vLLM, distributed/multi-GPU training

Post-training SFT, LoRA, distillation, model routing, evaluation

Systems HPC/SLURM, Kubernetes, Docker

Cloud AWS, GCP, Azure

PRIZES AND SCHOLARSHIPS

"la Caixa" foundation fellow (2022)

Excellence Scholarship to the best Spanish postgraduate students.

ELLIS PhD student (2022)

Coordinated by Prof. Ivan Titov (University of Edinburgh) and Prof. André F. T. Martins (Instituto Superior Técnico, University of Lisboa)

UKRI Centre for Doctoral Training in Natural Language Processing (2022)

Four year PhD program.

ACADEMIC DUTIES

Teaching Assistant in Foundations of Natural Language Processing (2022-2024)

Preparation of course materials: coursework, tutorials and laboratories. Weekly office hours. Answering online queries from students.

Reviewer for the ACL Rolling Review (ACL, EMNLP, NAACL) and NeurIPS